

SEEIR: Semantic driven End-to-end Exploration and Inspection for optimizing 3D Reconstruction Quality and Coverage*

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1st Oluwatosin Oseni
Robotics Department
Colorado School of Mines
Colorado, United States
oluwatosin_oseni@mines.edu

2nd Shengjie Wang
Institute for Interdisciplinary Information Sciences
Tsinghua University
Beijing, China
wangsj23@mails.tsinghua.edu.cn

3rd Dr. Micah Corah
Computer Science Department
Colorado School of Mines
Colorado, United States
micah.corah@mines.edu

Abstract—We present SEEIR, a semantic-driven end-to-end framework that jointly explores unknown environments and actively inspects a specified target object to maximize 3D reconstruction coverage and quality on an unmanned ground vehicle. The robot uses a navigation camera for safe exploration and a pan-tilt-zoom (PTZ) inspection camera for high-resolution observations. Exploration optimizes a joint objective that reduces map uncertainty while increasing surface-visibility coverage for efficient semantic target search. After target discovery, the agent transitions to active inspection and is rewarded with a bounded, diminishing-returns quality measure that directly leverages PTZ zoom by maximizing pixel-density alignment-based information gain over discretized mesh faces. We train a multimodal policy in photorealistic simulation using egocentric occupancy and visibility maps and a trajectory-based exploration bonus. We evaluate SEEIR by reconstructing a diverse set of objects using target point clouds and 3D Gaussian Splatting, and we compare reconstruction quality against an expert policy.

Index Terms—Reinforcement Learning, Active Perception, 3D Reconstruction, 3D Gaussian Splatting, Robot Inspection and Exploration.

I. INTRODUCTION

We propose a semantic-driven active perception algorithm that combines deep reinforcement learning (DRL) with classical information-gain exploration for robot inspection. We decouple the approach by mounting an actuated pan-tilt-zoom (PTZ) camera on an unmanned ground vehicle (UGV). The system uses an egocentric occupancy map for exploration and a surface-visibility map for coverage measurement. We train an agent to autonomously inspect a semantically unknown target in an unstructured, clustered warehouse environment. We also provide an open-source RL training environment that supports 128 concurrent environments.

Prior learning-based PTZ systems have been applied to active perception tasks mostly for object tracking and monitoring and surveillance [1] and more recent work apply Reinforcement learning to these tasks mostly to learn policies to actuate

the camera for high resolution tracking usually by zooming eliminating performance bottle necks with classic methods.

, but learning systems that combine semantic object search with high-quality inspection for 3D reconstruction on a UGV remain underexplored in the literature.

Intro outline reminder (move/remove later): Exploration; Object Search; Semantic Object Search; Inspection + Active 3D Reconstruction; PTZ Control for robotics.

A. Semantic Object Search

Prior work shows that semantic reasoning and common-sense priors can improve exploration efficiency by biasing search toward likely object locations [2]–[5]. However, many practical scenarios (e.g., cluttered environments or disaster response) lack reliable semantic context between the target and its surroundings; in these cases, the agent must perform more exhaustive search and typically uses semantic perception primarily as a detector to achieve the required object search/inspection objective [6]–[8].

B. SEEIR

At the start of each deployment, the robot is provided with an unknown semantic target. It performs an agnostic search because there is no context for where the object might be based on its type. The workflow begins with a joint process of semantic object search and exploration. Once the semantic target is found, the agent performs high-resolution 3D inspection optimized for reconstruction with 3D Gaussian Splatting through path planning, collision avoidance, and occlusion reasoning.

SEEIR is an agent equipped with a navigation camera for planning and collision avoidance, a PTZ camera for high-resolution inspection and 3D reconstruction, and IMU/odometry for state estimation.

II. PROBLEM FORMULATION

Our objective is to autonomously navigate an unknown environment to locate a 3D object with a semantic target ξ

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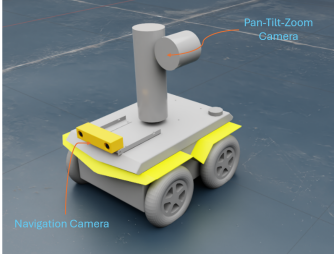


Fig. 1. Robot platform and sensor suite used for exploration and inspection.

and surface $\mathcal{S}$. We represent the surface $\mathcal{S}$ as a discrete set of observable faces $F = \{f_1, f_2, \dots, f_N\}$, where N is the total number of faces. The agent aims to maximize both the visual coverage of F via the cumulative inspection quality $Q(f_i) \in [0, 1]$ of each observed face f_i . We formulate the inspection task as a Partially Observable Markov Decision Process (POMDP).

- Action Space ($\mathcal{A}$): The action $a_t \in \mathbb{R}^5$ controls the base and the camera optics simultaneously:

$$a_t = [v_t, \omega_t, \dot{\theta}_{pan}, \dot{\theta}_{tilt}, \dot{f}_t]$$

Where v, ω are the differential drive linear and angular velocities, $\dot{\theta}$ represents the angular velocities of the camera joints, and $\dot{f}_t$ is the rate of change of the focal length (zoom).

- Observation Space ($\mathcal{O}_t$) We define our observation space obtained at every time step as

$$\mathcal{O}_t = [I_t, \Omega_t, \mathcal{M}_t, s_t] \quad (1)$$

I_t represents the RGBD Image information from the navigation and inspection cameras, Ω_t is a binary image obtained by semantic masking out our object of interest, and $\mathcal{M}_t$ encodes the environment's spatial information as maps. $s_t = [p_t, q_t, v_t, \omega_t, u_{t-1}, \theta_t^{pan}, \theta_t^{tilt}]$ represents the proprioceptive state of the robot of 3D position p_t , orientation q_t as 4D quaternions, 3D linear and angular velocity v_t, ω_t , controller actions u_{t-1} from previous time step, as well as the joint positions of the pan θ_t^{pan} and tilt articulations θ_t^{tilt} of our PTZ camera in radians.

III. ROBOT SETUP AND SENSOR SUITE

A. Navigation Camera

For navigation, we use a forward facing RGBD camera mounted on the robot. This camera depth data is used to generate our occupancy map, intended for navigating the unknown environment, and the depth data is used with ray tracing to generate our occupancy map

B. Pan-Tilt-Zoom (PTZ)/Inspection Camera

Attached to the top of the mobile base is an RGBD camera used to generate the Surface Visibility Map. This Camera is responsible for inspecting our inspection target ξ , and we run our semantic segmentation masking through this camera.

IV. ROBOT INSPECTION AND EXPLORATION

A. Environment Representation

To provide our agent with the spatial features of our environment, we provide our agent with a local egocentric map.

1) *Occupancy Channel* $M_{occ} \in [0, 1]$: Using the front Navigation Camera, we generate an Occupancy map, where each voxel encodes the probability of a cell being occupied. This provides the agent with spatial information for collision-free path planning to the goal. Where the value of a voxel v_i encodes the probability $p(v_{i,j,k})$ of a voxel being occupied

2) *Surface Visibility Channel*: $M_{vis} \in [0, 1]$: Using the Inspection camera attached to the robot's Pan-Tilt-Zoom articulation, we generate a surface visibility map via ray tracing, tracking only occupied cells. This provides the agent with a spatial representation of the surfaces observed.

3) *Visitation Map* M_{visit} : Using the robot's odometry, we maintain a 3D map that tracks the robot's trajectory. This map provides a curiosity bonus to encourage the robot to drive around the environment. The Map is updated by increasing the count of each voxel at every time step the voxel is occupied by the robot.

B. Exploration and Semantic Object Search

The robot's objective is divided into 3 distinct objectives: exploring and building a map of the environment, finding the semantic target ξ via surface coverage, and, finally, ensuring high-quality inspection of the target ξ via surface coverage.

1) *Robot Exploration and Entropy Reduction*: To enable our robot to find our inspection target ξ , we use an information-theoretic object to reduce the map's entropy. To encourage generalizable exploration policies and learning the spatial representation of the environment, we aim to minimize the robot's uncertainty about the map.

Using the Shannon entropy definition, we define the entropy of a voxel $v_{i,j,k}$ as the binary entropy of its occupancy probability $p_{i,j,k} = p$:

$$\mathcal{H}(v_{i,j,k}) = -[p \log(p) + (1 - p) \log(1 - p)] \quad (2)$$

Consequently, the total entropy of the map M , assuming independence between voxels, is the summation of the entropy of all voxels:

$$\mathcal{H}(M) = \sum_{i,j,k} \mathcal{H}(v_{i,j,k}) \quad (3)$$

2) *Surface Visibility Coverage*: The exploration described in Section IV-B1 is driven by observations from the navigation camera; however, this does not guarantee that we will observe the inspection target with our PTZ Camera. For this, we use an objective function that maximizes the information gain of the Surface Visibility map. To encourage map coverage, we use a saturation function for observing a surface. The current visibility score $\mathcal{S}(\mathcal{V}_t)$ is a weighted sum over the visibility map cells $v_{i,t}$:

$$\mathcal{S}(\mathcal{V}_t) = \sum_{i=1}^N -\gamma_v e^{-k_v v_{i,t}} \quad (4)$$

C. Inspection Quality Measure

Once the inspection target ξ is in view, the agent must maximise the inspection quality. We define the inspection quality $Q(f_i)$ of a face in Π , and we introduce a bounded function based on the resolution of the observed face. We define A_{f_i} as the information gain derived from face f_i , calculated as the cumulative weighted pixel count. To encourage efficient coverage of all surfaces of the inspection target, we model view quality with diminishing returns and adapt the submodular objectives proposed in [9], suitable for a single camera capable of zooming. The quality score $Q(f_i) \in [0, 1]$ is defined as:

$$q(f_i) = 1 - \exp\left(\frac{-A_{f_i} \cdot D_f}{\tau}\right) \quad (5)$$

Here, $D_f \in (0, 1]$ is a depth-based weighting term. Let d_f denote the camera-to-face distance, and let $d_{\max}$ be the desired maximum inspection distance. We define

$$D_f(d_f) = \exp\left(-\frac{\Delta d^2}{2\sigma_d^2}\right) \quad (6)$$

where σ_d controls how quickly the weight decays for observations farther than $d_{\max}$, and

$$\Delta d = \max(0, d_f - d_{\max}). \quad (7)$$

While [9] approximates density analytically using face distance, our system utilizes a PTZ camera to directly maximize visual resolution. We calculate the effective area A_f as the sum of the camera alignment score for every pixel p that currently falls on the face to encourage centering faces:

$$A_{f_i} = \sum_{p \in \mathcal{P}_f} (\hat{n}_f \cdot \hat{v}_p) \quad (8)$$

Where $\mathcal{P}_f$ is the set of all camera pixels observing face f_i in the current frame, $\hat{n}_{f_i}$ is the surface normal of the face, and $\hat{v}_p$ is the viewing vector for that specific pixel. This formulation naturally rewards the PTZ capabilities as the camera zooms in, increasing the cardinality of the set $|\mathcal{P}_f|$ and, in turn, the reward A_f finally we weigh this by the Depth distance from observation defined by

D. Joint Exploration Objective for Object Search

To explore the unknown environment and locate our inspection target ξ , we aim to learn a policy π that maximizes the joint objective of increasing visibility coverage while reducing map uncertainty. We define the total objective J as:

$$J(\pi) = \mathcal{S}(\mathcal{V}_t) - \lambda \mathcal{H}(\mathcal{M}) \quad (9)$$

Our goal is to find the optimal policy $\pi^* = \arg \max_{\pi} J(\pi)$.

V. METHODOLOGY

A. Learning Joint policy for Inspection and Navigation

1) *Reward Design*: The overall reward obtained at each time step t is provided below

$$R(s_t, a_t) = R_{\text{inspect}} + R_{\text{exploration}} + R_{\text{visibility}} + R_{\text{visitation}} - R_{\text{penalty}}$$

2) *Exploration Reward $R_{\text{exploration}}$* :

$$R_{\text{exploration}} = \gamma_E \mathcal{I}_t(\mathcal{M}) \quad (10)$$

$\mathcal{I}_t(\mathcal{M})$ represents the information gain of our occupancy map using the Shannon entropy. We define the information gain at time step t as

$$\mathcal{I}_t(\mathcal{M}) = \mathcal{H}(\mathcal{M}_{t-1}) - \mathcal{H}(\mathcal{M}_t)$$

Where the total map entropy $\mathcal{H}(\mathcal{M}_t)$ is defined in equation 3

$$\mathcal{H}(\mathcal{M}_t) = \sum_{i=1}^N \mathcal{H}_i(\ell_{i,t}) \quad (11)$$

3) *Visibility Reward $R_{\text{visibility}}$* : We define our visibility Reward as

$$R_{\text{visibility}} = \gamma_v \mathcal{V}_t \quad (12)$$

Where $\mathcal{V}_t$ represents the Visibility Increase Reward to encourage viewing as many Surfaces as possible. We reward viewing Surfaces, with the reward diminishing as we view the same surface.

Finally, we can define the full visibility increase reward $\mathcal{V}_t$ as the marginal gain over the surface visibility map.

$$\mathcal{V}_t = \mathcal{S}(\mathcal{V}_t) - \mathcal{S}(\mathcal{V}_{t-1})$$

4) *Inspection Reward R_{inspect}* : We compute the inspection reward R_{inspect} using a submodular marginal gain function in the inspection quality (5) across all faces observed [9]

$$R_{\text{inspect}} = \frac{\gamma_I}{N} \left(\sum_f \max(0, q(A_{f,t}) - q(A_f^{\text{best}})) \right) + \mathcal{C}_T$$

$\mathcal{C}_T$ is a terminal coverage bonus awarded at the end of an episode when the agent meets the current curriculum coverage threshold (i.e., achieves sufficient inspection coverage) and N is a normalisation factor which varies from semantic target and is equivalent to the number of unique faces of a surface of an object.

5) *Control Smoothing Penalty R_{penalty}* : To discourage jitter behavior in the PTZ Camera, we apply a penalty for large changes in the control values. We compute this penalty as the squared L_2 norm of the temporal difference between the consecutive PTZ control vectors:

$$R_{\text{penalty}} = \|\mathbf{u}_{t,\text{ptz}} - \mathbf{u}_{t-1,\text{ptz}}\|_2^2 \quad (13)$$

where $\mathbf{u}_{t,\text{ptz}} = [u_{\text{pan}}, u_{\text{tilt}}, u_{\text{zoom}}]^T$ represents the PTZ control value at time t . Additionally, a constant time-step penalty is applied to encourage the agent to complete the inspection task efficiently.

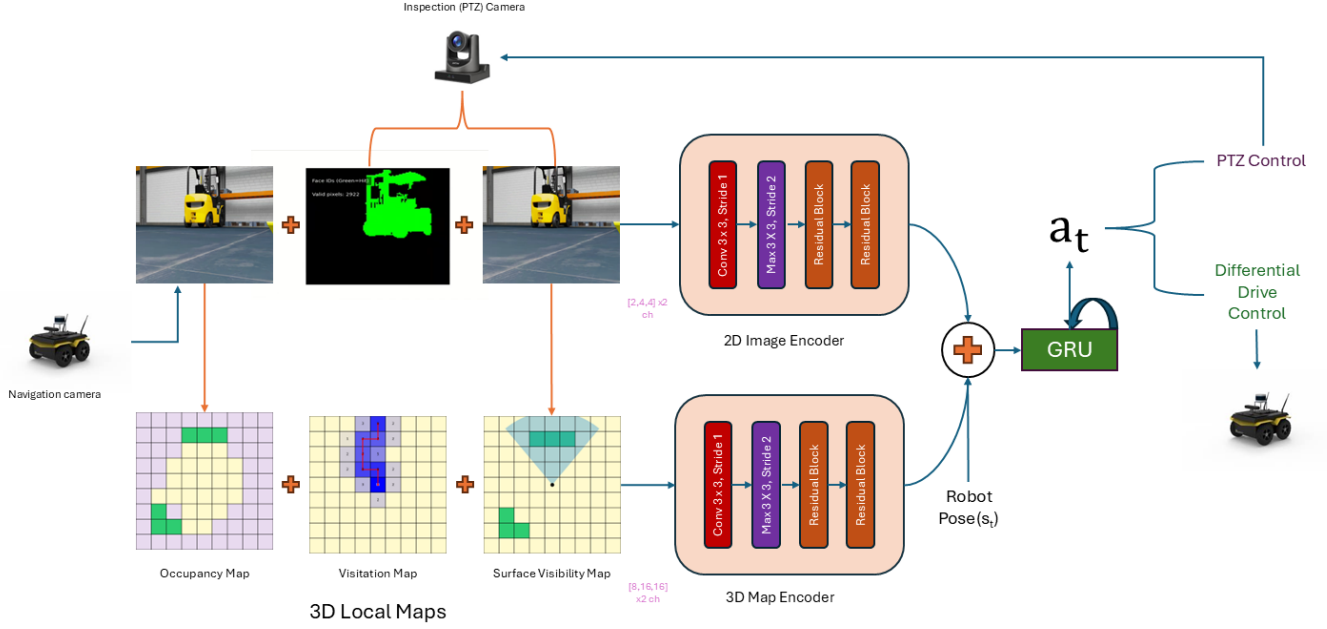


Fig. 2. System Architecture

B. Neural Network Architecture and inputs

Our agent processes multi-modal inputs designed for spatial and temporal data.

1) *2D Visual Inputs*: The RGB data from the Inspection, RGB-D from the Navigation camera, as well the binary Semantic image are stacked and passed through the 2D ResNet feature extractor (Image Encoder)

2) *3D Spatial Mapping*: To provide our agent with local spatial information, we utilize an egocentric local $\mathcal{M}_t \in [n, n, \frac{n}{2}, c]$ centered around the robot, where $n = 2.1$ meters with a resolution $r = 0.1$ and c represents the 3 channels (Occupancy, Surface Visibility, Visitation). This map is processed via a 3D ResNet feature encoder (Map Encoder)

C. Modality Embedding and Fusion

To effectively fuse the distinct observation spaces, we employ a multimodal Transformer architecture [10], [11]. First, the high-dimensional camera features, local map features, and low-dimensional proprioceptive robot pose are independently linearly projected into a shared latent dimension to form individual modality tokens. To stabilize training, these tokens are normalized via Layer Normalization before learnable modality embeddings are added to retain the identity of each sensor source. The resulting token sequence is then processed by a multi-layer self-attention Transformer encoder. This architecture allows the model to capture rich cross-modality dependencies, enabling each modality to contextualize and conditionally attend to features from the others. Finally, we apply average pooling across the output token sequence to produce a single, unified feature representation that serves as the fused embedding.

D. Temporal Processing

This vector from ?? is fed into a GRU (Gated Recurrent Unit) network with a history length of 32 previous states to encode temporal information. The GRU encodes temporal information into the M-D feature, which we finally feed to our Policy and Value MLP networks.

VI. IMPLEMENTATION

We train our policy using a slightly modified PPO-GRU (Proximal Policy Optimization; [12]) implementation from the SKRL reinforcement learning library [13]. To encode multi-modal observations—local occupancy and visibility maps, as well as RGB and semantic images from the navigation and inspection (PTZ) cameras—we use the 3D/2D ResNet encoder from IMPALA [14].

VII. TRAINING ENVIRONMENT AND SIMULATION

This section describes the simulation setup used for training and evaluation. Training is performed on an NVIDIA RTX 8000 using Isaac Lab [15] in the open-room Warehouse environment from the Isaac Lab Assets.

We use a modified Clearpath Jackal by adding a custom PTZ articulation and increasing the control space to 7 DoF: 4 wheel joints, 2 joints for pan/tilt, and 1 control input for camera zoom. The inspection camera focal length is constrained to $f_t \in [10, 100]$ ($10\times$ zoom) simulating the real world capabilities of PTZ Cameras.

The navigation camera and the inspection (PTZ) camera are implemented as tiled RGB-D cameras; for face/mesh surface identification we use a ray-caster camera, which provides per-face hit information on discretized mesh geometry.

Each episode is limited to 60 s. The Jackal is constrained to a maximum linear velocity of 2 m/s and an angular velocity of 4 rad/s. Using GPU-accelerated ray tracing and depth from the navigation and inspection cameras, we maintain (i) a global occupancy map and (ii) a surface-visibility map for each episode. The map bounds are $X \in [10.5, 9.5]$, $Y \in [12.5, 18.0]$, and $Z \in [0.0, 2.5]$ (with the Z range constrained by ground-vehicle motion), using a resolution of 0.25 m/voxel. We update the maps at 5.4 Hz and use a physics simulation rate of 130 Hz.

For policy inputs, we extract an agent-centric local map from the global maps to improve generalization and reduce sensitivity to global drift. We also inject uniform Gaussian noise into the proprioceptive state s_t during training.

VIII. EXPERIMENTATION AND ABLATION STUDIES

A. Experimental Setup

We evaluate SEEIR in the Isaac Lab Warehouse environment described in Section VII, with randomized robot and target initializations.

We add additional constraints to the scene using basic primitives (cones, spheres, etc.) with randomized positions and properties. Using a 2D grid, we procedurally generate a scene by uniformly sampling positions across the 2D warehouse space for constraints, inspection targets, and robots. To encourage generalization, we also randomize the inspected object category. To quantify reconstruction quality, we compare against ground truth using point-cloud metrics (e.g., Chamfer distance, Hausdorff distance, and coverage at a distance threshold τ). We also report 3D Gaussian Splatting quality using PSNR where applicable.

B. Ablation Study

C. Simulation Studies

We compare our Method to pure exploration information-theoretic methods. First with the same input states as SEEIR we perform pure exploration with

D. Results

We compare against a fixed-trajectory baseline that drives the Jackal around the inspection target with a fixed focal length. We evaluate the reconstructed point cloud against a ground-truth point cloud obtained by sampling the object mesh surface. We report a symmetric distance (Chamfer) for overall geometric error, a worst-case distance (Hausdorff), and completeness as coverage (percentage of ground-truth points within a distance threshold τ of the reconstruction).

IX. REAL-WORLD DEPLOYMENT

X. FUTURE WORK

Our future work involves deploying our agent on a Real-World Jackal robot and Pan Tilt Zoom Camera. Towards the goal of building a more generalist agent we aim to prompt the agent with Natural Language

TABLE I
RECONSTRUCTION QUALITY BETWEEN GROUND TRUTH, FIXED TRAJECTORY AND SEEIR

Metric	Fixed Trajectory	SEEIR
Chamfer Distance (mean)	0.0632	
Hausdorff Distance	0.3482	
Coverage @ $\tau = 0.01$	63.75%	
Coverage @ $\tau = 0.20$	94.17%	

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The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

- [1] S. S. Sandha *et al.*, “Eagle: End-to-end deep reinforcement learning based autonomous control of ptz cameras,” in *Proceedings of the 8th ACM/IEEE Conference on Internet of Things Design and Implementation*, 2023, pp. 144–157.
- [2] M. F. Ginting *et al.*, “Seek: Semantic reasoning for object goal navigation in real world inspection tasks,” *arXiv preprint arXiv:2405.09822*, 2024.
- [3] M. Dharmadhikari and K. Alexis, “Semantics-aware predictive inspection path planning,” *IEEE Transactions on Field Robotics*, 2025.
- [4] C. Wang *et al.*, “Semantic-aware informative path planning for efficient object search using mobile robot,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 8, pp. 5230–5243, 2019.
- [5] K. Nagami *et al.*, “Vista: Open-vocabulary, task-relevant robot exploration with online semantic gaussian splatting,” *IEEE Robotics and Automation Letters*, 2026.
- [6] G. Malczyk, M. Kulkarni, and K. Alexis, “Semantically-driven deep reinforcement learning for inspection path planning,” *IEEE Robotics and Automation Letters*, 2025.
- [7] M. Dharmadhikari and K. Alexis, “Semantics-aware exploration and inspection path planning,” *arXiv preprint arXiv:2303.07236*, 2023.
- [8] T. Dang *et al.*, “Autonomous search for underground mine rescue using aerial robots,” in *2020 IEEE Aerospace Conference*. IEEE, 2020, pp. 1–8.
- [9] S. Hughes *et al.*, “Multi-robot planning for filming groups of moving actors leveraging submodularity and pixel density,” in *2024 IEEE 63rd Conference on Decision and Control (CDC)*. IEEE, 2024, pp. 5401–5408.
- [10] R. Bachmann *et al.*, “MultiMAE: Multi-modal multi-task masked autoencoders,” 2022.
- [11] O. M. Team *et al.*, “Octo: An open-source generalist robot policy,” *arXiv preprint arXiv:2405.12213*, 2024.
- [12] J. Schulman *et al.*, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [13] A. Serrano-Muñoz *et al.*, “skrl: Modular and flexible library for reinforcement learning,” *Journal of Machine Learning Research*, vol. 24, no. 254, pp. 1–9, 2023. [Online]. Available: <http://jmlr.org/papers/v24/23-0112.html>
- [14] L. Espeholt *et al.*, “Impala: Scalable distributed deep-rl with importance weighted actor-learner architectures,” in *International conference on machine learning*. PMLR, 2018, pp. 1407–1416.
- [15] M. Mittal *et al.*, “Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning,” *arXiv preprint arXiv:2511.04831*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.04831>